Learn basic GIS

GIS and QGIS

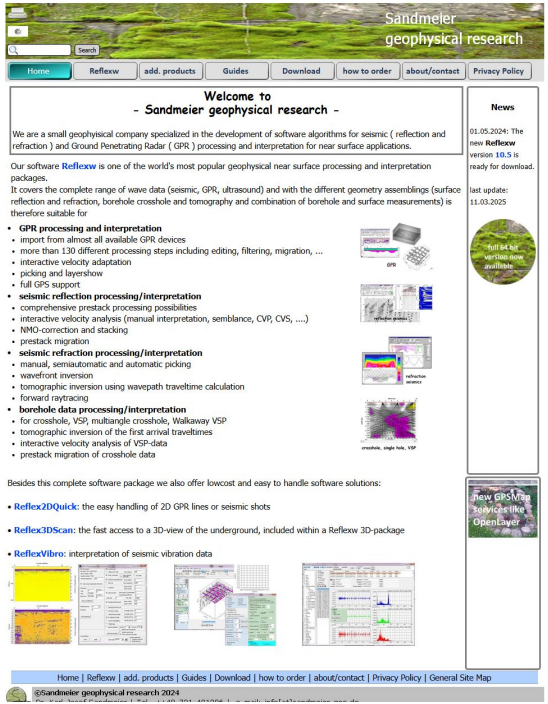
GIS (Geographic Information System) is a technology used to visualize, analyze, and manage spatial or geographic data.

QGIS is a free and open-source GIS software that allows us to create maps, analyze spatial data, and represent georeferenced information.

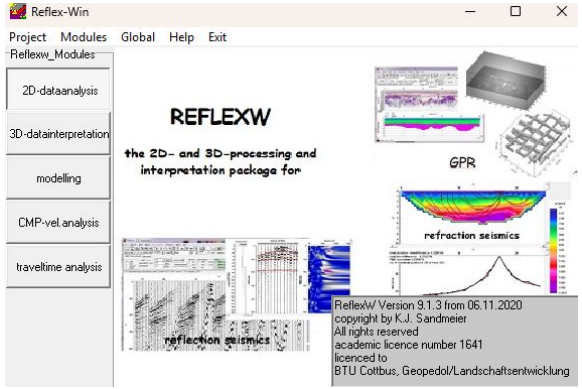
We are going to use QGIS to:

- Load GPS points
- Represent measurement profiles (e.g., from fieldwork or lab)
- Visualize and export maps with scale bars, north arrows, and labels
- Combine spatial data with analysis tools

Reflexw



Manual and Videos →
https://www.sandmeier-geo.de/index.html



Our guide →
https://docs.google.com/document/d/1-pYSXHyyq7VpLRZ5x-Jyaaif0vGlBCSb9d42LRsN3Jw/edit?usp=sharing

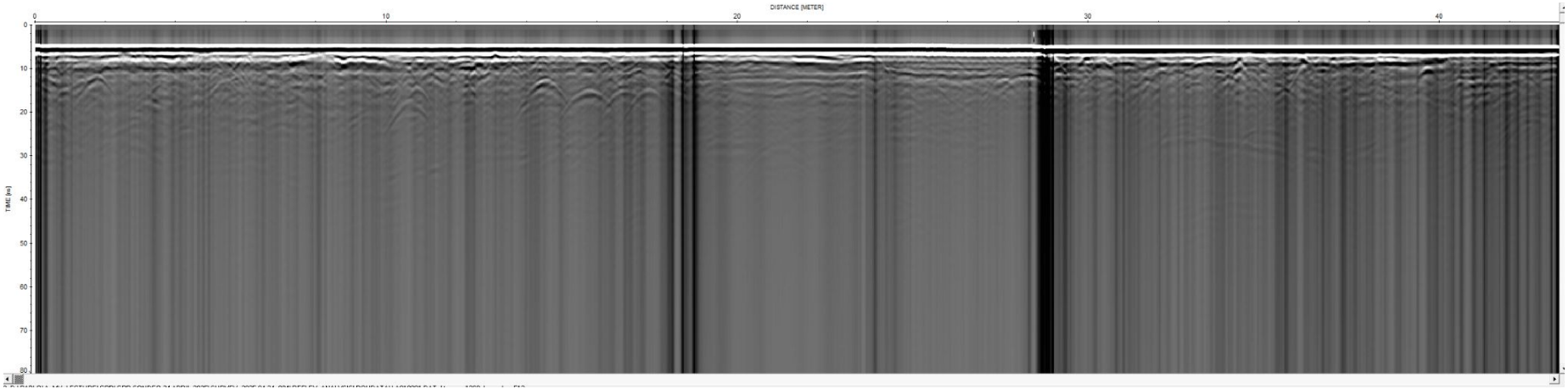
Data processing

1) **Get familiar with ReflexW**: basic steps include running HD Converter, creating project folders, loading data, importing profiles. Then explore the basics of *plot options*.

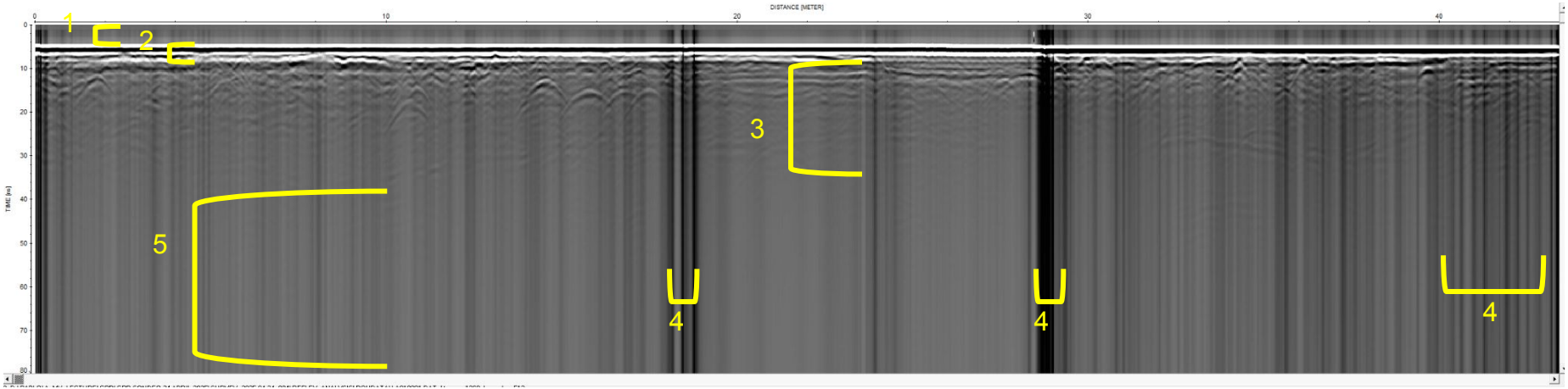
2) **Apply filters**: progressively apply processing filters to remove noise and enhance reflections.

3) **Initial visual analysis**: look at the profile globally, identify major features, distinguish between the two frequencies, and compare with the map generated in the field.

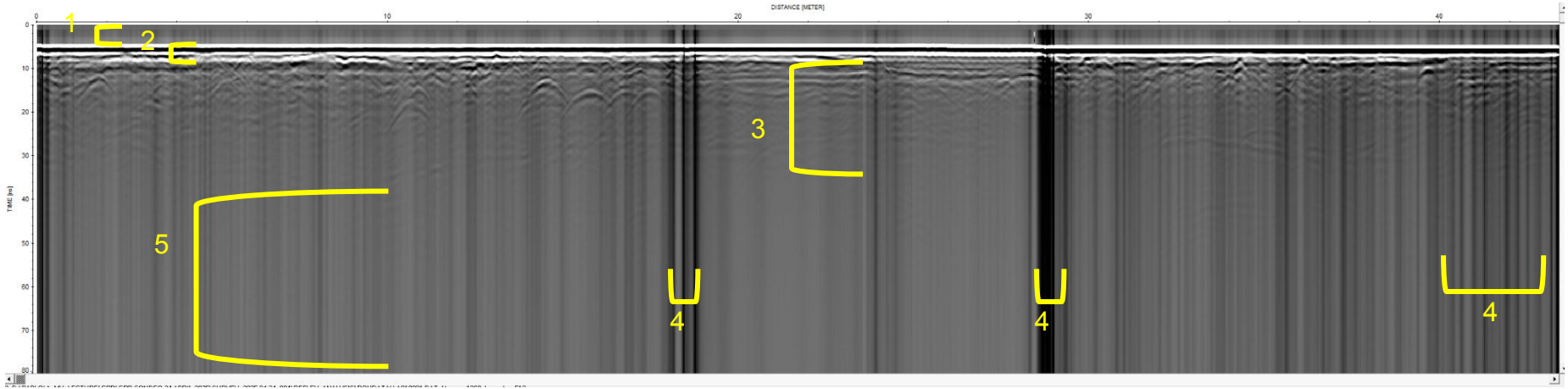
What we see from the start…



What we see from the start…

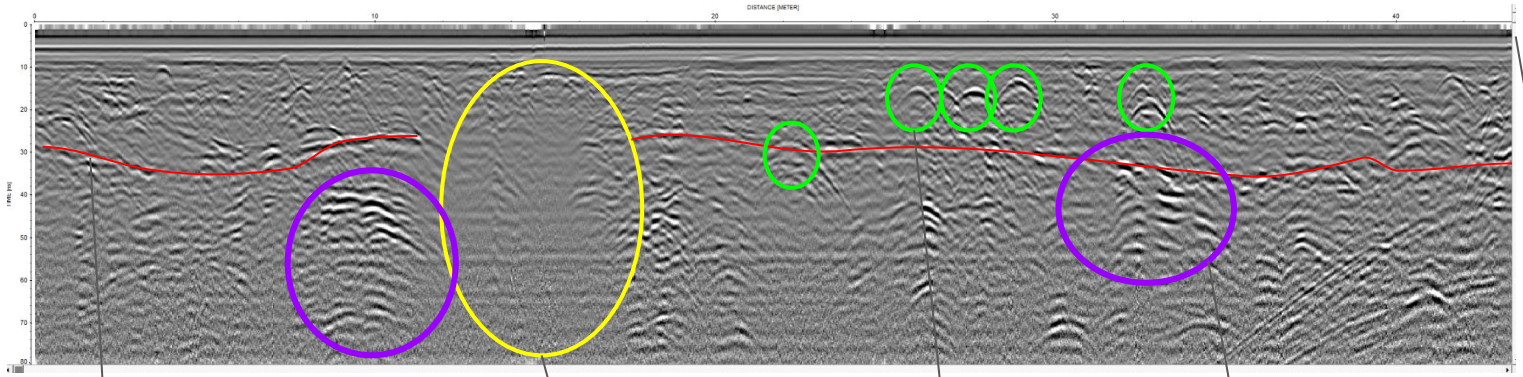


What we see from the start…



1) **Pre direct wave zone** → air gap or time before any signal arrives
2) **DIrect wave** → Signal travelling directly from the transmitter to the receiver → remove with background removal or muting
3) **Subsurface reflections** → interfaces layers, buried objects (pipes rocks etc)
4) **Vertical noise** → this is cahused by adquisition issues and or hardware interference. These are not real subsurface objects → spike removal
5) **Deep noise** → difffuse and irregular reflections; caused by signal attenuation

And after the processing…



interface:
fill to natural soil?

Homogeneous or low constrast material:
Maybe compacted fill without objects?
or uniform dry sand?

Point objects, buried structures:
pipes, rocks, metallic elements

construction remains?
metallic elements?
compacted material?

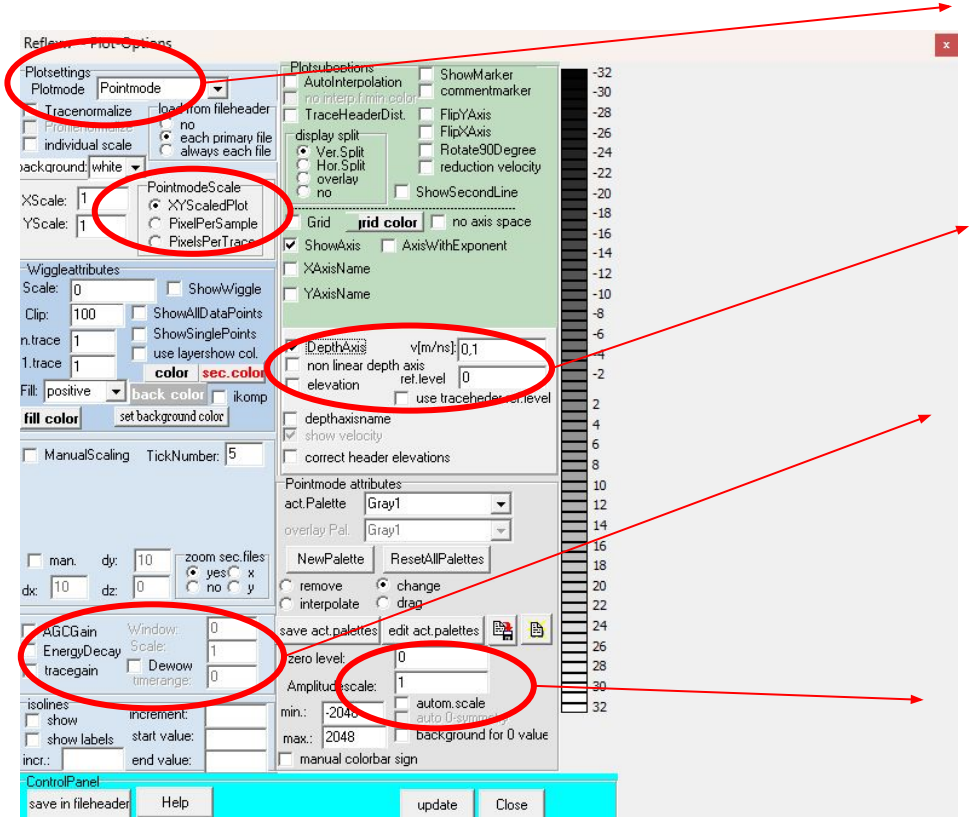
Direct signal and contact with pavement

Review main concepts from previous class

Start working with the data.

(You can follow the guide)

Getting familiar with ReflexW:



Plotsettings:

- Choose between Wiggle mode (traces as wiggles) and Radargram mode (amplitude as grayscale).

Depth Axis:

- Set the propagation velocity (v) to convert time (ns) to depth (m). Adjusting the velocity according to the subsurface material.

Basic Filters:

- Energy Decay: compensation for the loss of signal energy with depth. We use it for visualizing deeper reflections
- Dewow: Removes low-frequency noise. Essential for getting clearer reflections.

Amplitude Scale:

- Adjusts the visual intensity of the traces : enhance weak reflections or avoid saturation

Depth Axis

- The velocity value we are going to enter in ReflexW for DepthAxis is used just to convert the vertical time axis to depth.
- If there are multiple materials, it is common to use an average value representative of the dominant material.

$$propagation\ speed = \frac{c}{\sqrt{\varepsilon_r}}$$

$$depth = \frac{velocity * time}{} = \frac{propagation\ speed * range}{2} = \frac{c}{\sqrt{\varepsilon_r}} * \frac{range}{2}$$

$$c = 30\ cm/ns$$

- Air $\epsilon_r = 1\ v \simeq 30\ cm/ns$
- Dry Sand $\epsilon_r = 3\ to\ 6\ \ v \simeq 12.2\ cm/ns\ to\ 17.3\ cm/ns$
- Wet Sand $\epsilon_r = 20\ to\ 30\ \ v \simeq 5.5\ cm/ns\ to\ 6.7\ cm/ns$
- Dry Soil $\epsilon_r = 4\ to\ 6\ \ v \simeq 12.2\ cm/ns\ to\ 15\ cm/ns$
- Wet Soil $\epsilon_r = 10\ to\ 15\ \ v \simeq 7.7\ cm/ns\ to\ 9.5\ cm/ns$
- Fresh Water $\epsilon_r = 80\ \ v \simeq 3.4\ cm/ns\ to$
- Concrete $\epsilon_r = 6\ to\ 12\ \ v \simeq 8.7\ cm/ns\ to\ 12.2\ cm/ns$
- Asphalt $\epsilon_r = 4\ to\ 10\ \ v \simeq 9.5\ cm/ns\ to\ 15\ cm/ns$

Setting an incorrect speed results in depth errors:

- Too high speed: Depth is overestimated
- Too low speed: Depth is underestimated

Dewow Filter: Removing Low-Frequency Noise

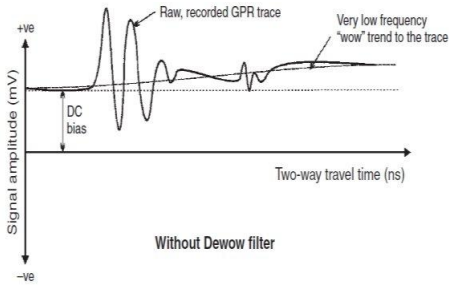
The filter is used to eliminate very low-frequency components that are not part of the actual reflections.

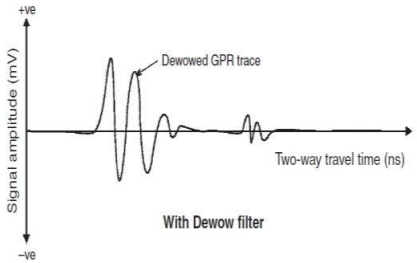
Why do we need to filter

1. **Noise from the equipment (DC noise):**
   - Electronic systems generate slow baseline drift and low-frequency noise.
   - These components appear as long, smooth oscillations or a curved baseline in the radargram.

2. **Low-frequency components within the pulse spectrum:**
   - A 200 MHz antenna does not emit a pure 200 MHz tone but a broadband pulse.
   - The emitted signal contains spectral components both below and above 200 MHz.
   - Dewow may remove very low-frequency parts of the pulse spectrum, and it wont affect useful signals if the window size is appropriate.

$$f_c = \frac{1}{2T}$$

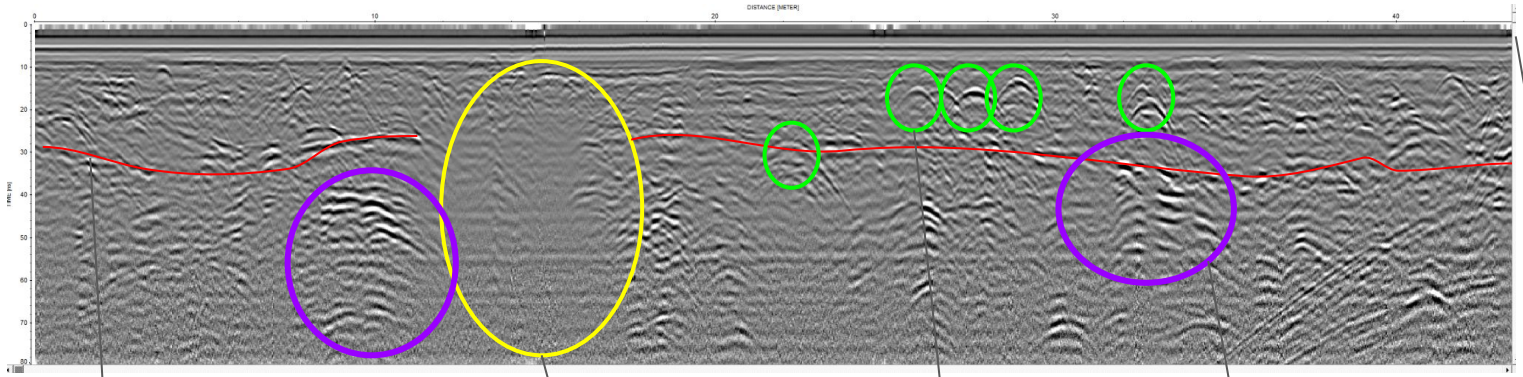




Antenna freq	Dewow time window (aprox)	Cut off freq (aprox)
200 MHz	10-20 ns	50-25 MHz
600 MHz	2-5 ns	250-100 MHz

Interpretation

600 Mhz antenna - just after modifying dewow, gain, and plot scale



interface:
fill to natural soil?

Homogeneous or low constrast material:
Maybe compacted fill without objects?
or uniform dry sand?

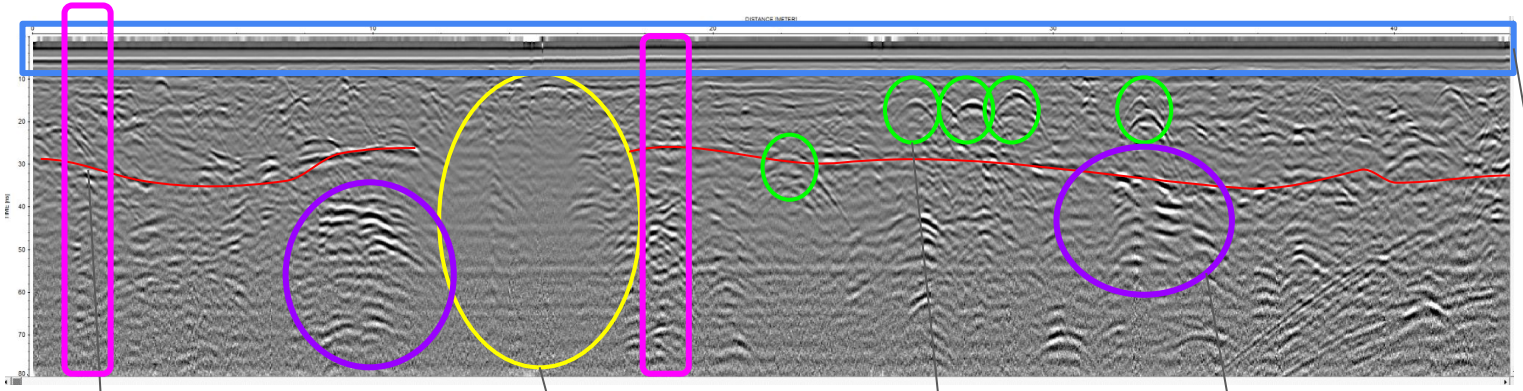
Point objects, buried structures: pipes, rocks, metallic elements

construction remains?
metallic elements?
compacted material?

Direct signal and contact with pavement

Interpretation

600 Mhz antenna - just after modifying dewow, gain, and plot scale



interface:
fill to natural soil?

Homogeneous or low constrast material:
Maybe compacted fill without objects?
or uniform dry sand?

Point objects, buried structures:
pipes, rocks, metallic elements

construction remains?
metallic elements?
compacted material?

Direct signal and contact with pavement

Data processing

Data import and initial setup

- Load raw data and se correct parameters (units, time window)

Basic processing

- Flip (in case is necessary)
- Static correction: Align the time zero to account fo signal delay

Amplitude and gain adjustments

- Amplitude scale. You can adjust it for optimal visual contrast
- Energy decay. This compensates for amplitude loss

Noise reduction and filtering

- Background removal. This subtracts average trace to eliminate systematic noise
- Band pass filter. Keep only the frequencies you want within a useful range.

Advanced processing

- Migration. this corrects hyperbolic reflection to their true position
- Deconvolution. Increase resolution by compressing the wavelet.

DO NOT overprocess the data!